Abdul Jaweed

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PROFILE

AI Engineer with 3+ years of experience designing and building enterprise AI applications, knowledge graph systems, and production-grade AI platforms. Experienced in **Knowledge Graph Engineering, ontology and taxonomy modeling, graph topology, entity-relationship modeling, data granularity, and graph-based knowledge representation** using Neo4j, Cypher, Graph Data Science (GDS), and APOC. Specialized in **GraphRAG, graph-based memory systems, LLM-powered applications, AI agents, and Retrieval-Augmented Generation (RAG)** using Python, FastAPI, LangGraph, and modern AI frameworks. Skilled in integrating structured and unstructured data with LLMs, developing secure and scalable APIs, and building multi-tenant AI systems with authentication, authorization, observability, and production-grade backend infrastructure. Passionate about engineering reliable, scalable, and maintainable AI solutions that transform complex data into actionable intelligence.

SKILLS

Programming Languages: Python, SQL, Pandas, NumPy, data preprocessing, text cleaning
Machine Learning & NLP: scikit-learn, classification, regression, feature engineering, model evaluation, NLP, transformer models, text summarization
GenAI & LLM Engineering: LLM integration, prompt engineering, semantic search, hybrid search, retrieval/reranking, vector search, context engineering, embeddings, RAG, GraphRAG, semantic retrieval, conversational AI, agent evaluation.
Agents & Orchestration: LangGraph, LangChain,Agentic architecture, multi-agent systems, tool/function calling, Model Context Protocol (MCP), SSE streaming, workflow orchestration
Knowledge Graphs & Semantic Engineering: Knowledge Graphs, Neo4j, Cypher, Graph Data Modeling, Taxonomy, Ontology, Semantic Modeling, Semantic Enrichment
Backend & API Engineering: FastAPI, REST APIs, Server-Sent Events (SSE), Pydantic, SQLAlchemy, Alembic, Celery
Databases & Data Engineering: PostgreSQL, Redis, MongoDB, pgvector, Qdrant, S3, Pandas, NumPy.
Identity, Security & API Management: ZITADEL, Cerbos, Tyk API Gateway, OAuth 2.0, OpenID Connect (OIDC), JWT Authentication, Role-Based Access Control (RBAC), Policy-as-Code, Row-Level Security (RLS), API Security.
Observability & Monitoring: OpenTelemetry, Langfuse, Vector, Structured Logging, Metrics and Traces.
DevOps & MLOps: Docker, Docker Compose, Git, GitHub Actions, Railway Cloud, Experiment Tracking, Model Deployment.
AWS Cloud & Infrastructure:IAM, S3, EC2, ECR, Bedrock, Lambda, VPC, Subnets, NAT Gateway, Internet Gateway, Route Tables, ALB, WAF, VPC Peering, Transit Gateway, API Gateway.
LLM-Finetuning: LoRA, QLoRA, Distillation, RLHF, DPO
AI Coding & Agentic Development: Cursor, Claude Code, Codex, OpenCode, Antigravity, Hermes Agent

MY PERSONAL PROJECTS TURN INTO OPEN SOURCE CONTRIBUTIONS

DataHek OSS — Universal Conversational Data Platform

Status: Active Development
DataHek-OSS is an open-source agentic platform that converts natural-language questions into validated, read-only SQL queries against live databases. It combines LLM-based planning with deterministic validation, guardrails, auditing, data masking, and real-time streaming.
Tech Used: Python, FastAPI, MCP, Agent Skills, Multi-Agent Systems, Neo4j, Knowledge Graphs, PostgreSQL, Redis, Infisical

- Built the end-to-end agent pipeline covering schema discovery, LLM planning, deterministic validation, guardrails, auditing, masking, streaming, and retry recovery.
- Built a **Neo4j schema knowledge graph** using taxonomy, ontology, topology, and granularity to enrich agent context and improve SQL reasoning.
- Developed provider-agnostic **Logical Query Plans (LQP)** with ClickHouse and PostgreSQL connectors and dialect-specific validation.
- Implemented real-time **SSE streaming** with live pipeline progress across connection, planning, execution, and explanation stages.
- Implemented **SSE streaming, PostgreSQL persistence, Redis configuration, and Infisical secret management**.

MLFAST

MLFAST — Python Machine Learning Library

Creator & Developer | Python, Scikit-learn, GitHub Actions, CircleCI

- Developed and published an open-source Python ML library on PyPI, simplifying regression and classification workflows through a high-level API.
- Integrated multiple machine learning algorithms, including linear regression, decision trees, random forests, SVM, and gradient boosting.
- Implemented configurable feature scaling using StandardScaler and RobustScaler.
- Added categorical feature handling through one-hot encoding.
- Enabled hyperparameter optimization to support model configuration and experimentation.
- Designed reusable interfaces to reduce boilerplate code and simplify model development.

MLFast-OSS — Unified MLOps Platform (OpenSource Soon)

Creator & Lead Engineer | Python, FastAPI, PostgreSQL, Redis, RustFS/S3, Celery, Next.js, Docker, Kubernetes, Terraform, GitHub Actions

Designed and built an end-to-end, multi-tenant MLOps platform that unifies the machine learning lifecycle—from dataset management and model training to evaluation, deployment, monitoring, and automated retraining—through a consistent API, Python SDK, CLI, and web interface.

- **End-to-End ML Lifecycle:** Architected versioned dataset management with checksums, experiment tracking, model training and hyperparameter optimization, evaluation quality gates, model registry with approval workflows, versioned pipeline DAGs, model serving, drift monitoring, and policy-driven automated retraining.
- **Unified Platform Architecture:** Developed a typed Python SDK, CLI, and Next.js control plane around a shared resource model, enabling consistent ML workflows across programmatic, command-line, and web interfaces.
- **CI/CD & Automated Quality Gates:** Implemented GitHub Actions pipelines for linting, type checking, automated testing, API contract validation, generated TypeScript types, infrastructure validation, Kubernetes integration tests, software bill of materials (SBOM), dependency security scanning, and frontend bundle budgets.
- **Testing & Reliability:** Established 1,729 automated tests across the backend and SDK, 14 Playwright end-to-end specifications, and 41 database schema migrations with automated migration-safety checks.
- **Security & Governance:** Implemented multi-tenant authentication and authorization using API keys, JWT, RBAC, quotas, versioned policy management with simulation, tamper-evident audit trails, and production configuration checks to prevent insecure deployments.
- **Deployment & Operations:** Packaged the platform using Docker Compose with PostgreSQL, Redis, S3-compatible RustFS, and Celery workers; added Kubernetes manifests, Terraform infrastructure definitions, and disaster-recovery tooling with tested backup and restore procedures.

Tech Stack: Python, FastAPI, PostgreSQL, Redis, Celery, RustFS/S3, Next.js, Docker, Kubernetes, Terraform, GitHub Actions, Playwright.

ACHIEVEMENT

Machine Learning Hackathon 02/2023

Achieved selection as a data science intern after successfully competing against over 7000+ participants in an ML hackathon and assessment test, ranking among the top 10 percent. Hyderabad, India

PROFESSIONAL EXPERIENCE

Human Managed. AI Engineer 11/2024 – 09/2026

Singapore

- Built a production AI security assistant using Python, FastAPI, SSE, MCP, AWS Bedrock, and LiteLLM for grounded, explainable enterprise responses.
- Designed **agentic AI architectures and multi-agent workflows** using LangGraph, tool orchestration, conversational memory, and context management.
- Integrated **Model Context Protocol (MCP) servers** with agentic AI workflows to enable standardized tool integration and contextual access to enterprise capabilities.
- Integrated **Azure OpenAI models and Azure AI Services** into LLM-powered applications for model inference, prompt orchestration, and enterprise AI workflows.
- Implemented six-layer AI guardrails covering input classification, risk scoring, tool sanitization, prompt hardening, output validation, and entitlement checks.
- Engineered Redis-backed conversational memory with server-side context assembly and token-budget controls for optimized latency and inference costs.
- Designed a Memory-as-a-Service platform using FastAPI, PostgreSQL/pgvector, Qdrant, and Redis for reusable short- and long-term agent memory.
- Developed **agentic AI and multi-agent workflows** using LangGraph, tool integration, and checkpointing to support orchestrated AI tasks.

- Built pluggable Knowledge Graph Memory using **Neo4j, Cypher/OpenCypher, and APOC**, with graph data modeling, taxonomy, ontology, entity resolution, and GraphRAG.
- Optimized **Neo4j graph database** workloads using **Cypher query optimization, indexing, traversal optimization, and APOC procedures**.
- Implemented enterprise security and reliability using RLS, JWT/API keys, JSON Schema validation, quotas, retention workflows, and LangGraph checkpointing.
- Delivered enterprise RAG and asset-intelligence platforms enabling grounded knowledge access, asset discovery, and graph-based relationship analysis.
- Supported **AI deployment and production operations** through containerized services, API integrations, monitoring, and observability.
- Supported **Azure-based and AWS-based AI deployment** by integrating model endpoints with backend APIs, configuration management, authentication, and production application workflows.

Capital Placement. Junior AI Engineer 11/2023 – 10/2024
United Kingdom

- Built AI-powered **candidate recommendation and screening workflows** using OpenAI APIs and open-source LLMs, including Llama 3, Mistral, Phi-3, and Gemma, incorporating prompt engineering and structured outputs.
- Developed data preprocessing and text-cleaning pipelines for **job descriptions and candidate profiles** to support AI-driven candidate matching.
- Developed backend APIs and integrated LLM capabilities into **HR technology workflows**, including tool/function calling and AI application integrations.
- Implemented **agentic AI and multi-agent workflow patterns** for candidate screening and interview automation, where applicable.
- Deployed AI services using **Azure Functions**, supporting cloud-based AI application delivery and deployment workflows.
- Built an **AI-powered interviewer** that conducted automated 10-minute candidate interviews and supported candidate screening without human intervention.
- Automated candidate shortlisting and interview-based screening workflows to improve recruitment efficiency.
- Explored **MCP integration and AI orchestration frameworks**, including LangChain and LlamaIndex, to support extensible LLM application architectures, where implemented.

Margdarshan. Data Scientist 06/2023 – 11/2023
India

- Built a **Generative AI Q&A chatbot** using OpenAI and Hugging Face APIs, enabling natural-language question answering and conversational AI capabilities.
- Implemented an **NLP text-summarization pipeline** using Hugging Face's facebook/bart-large-cnn transformer model and integrated CI/CD automation for repeatable delivery.
- Evaluated and refined Q&A and summarization workflows through testing to improve response quality, reliability, and user experience.
- Explored **agentic AI workflows, multi-agent orchestration, and MCP integration** to support extensible AI application architectures.
- Supported **AI application deployment** and CI/CD workflows, focusing on reliable delivery and maintainable backend integrations.
- Collaborated with cross-functional stakeholders to translate business requirements into AI-powered solutions aligned with project objectives.

COURSES

Machine Learning Specialization , *Deeplearning.ai from Coursera* 07/2022 – 10/2022

MLOps Specialization , *Duke University from Coursera* 06/2023 – 08/2023

EDUCATION

Bachelor of Commerce in Computer Applications, *Government Degree College Khairatabad* 06/2019 – 07/2022
Hyderabad, India

- **Cummulative CGPA : 7.82/ 10**